

HEX BUFFERS/INVERTERS**8T95, 96, 97, 98****High Speed Hex 3-State Buffers
High Speed Hex 3-State Inverters****DESCRIPTION**

Each of the 3-state bus interface elements described herein has low current PNP inputs and is designed with Schottky TTL technology for ultra high speed. The devices are used to convert TTL/DTL or MOS/CMOS to 3-state TTL bus levels. For maximum systems flexibility, the 8T95 and 8T97 do so without logic inversion, whereas the 8T96 and 8T98 provide the logical complement of the input. The 8T95 and 8T96 feature a common control line for all six devices, whereas the 8T97 and 8T98 have control lines for four devices from one input and two from another input.

TYPE	TYPICAL PROPAGATION DELAY	TYPICAL SUPPLY CURRENT (Total)
N8T95	8ns	65mA
N8T96	6.5ns	59mA
N8T97	8ns	65mA
N8T98	6.5ns	59mA

ORDERING CODE

PACKAGES	COMMERCIAL RANGES $V_{CC} = 5V \pm 5\%$; $T_A = 0^\circ C$ to $+70^\circ C$	MILITARY RANGES $V_{CC} = 5V \pm 10\%$; $T_A = -55^\circ C$ to $+125^\circ C$
Plastic DIP	N8T95N • N8T96N N8T97N • N8T98N	
Ceramic DIP	N8T95F • N8T96F N8T97F • N8T98F	S8T95F • S8T98F S8T97F •

FUNCTION TABLE—8T95

INPUTS			OUTPUT
DIS ₁	DIS ₂	I	Y
L	L	L	L
L	L	H	H
X	H	X	(Z)
H	X	X	(Z)

L = LOW voltage level
H = HIGH voltage level
X = Don't care
(Z) = HIGH impedance (off) state

INPUT AND OUTPUT LOADING AND FAN-OUT TABLE

PINS	DESCRIPTION	8T
DIS	Input	1Sul
I	Input	1Sul
Y	Output	24Sul

NOTE
A 54/74S unit load (ul) is $50\mu A$ I_{IH} and $-2.0mA$ I_{IL} .

FUNCTION TABLE—8T96

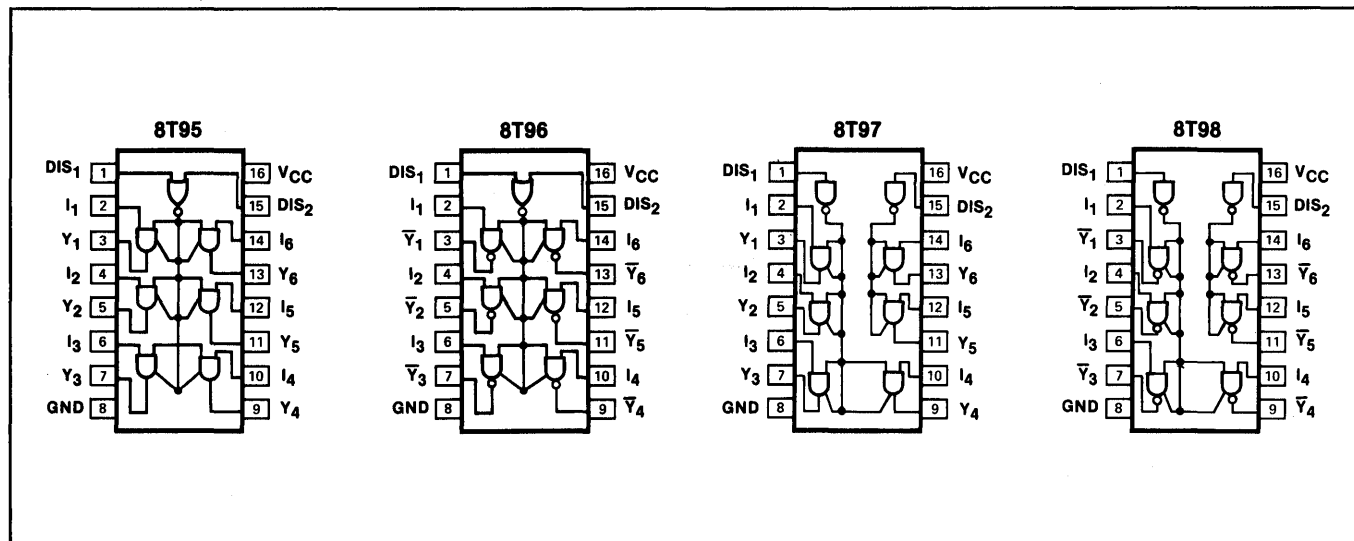
INPUTS			OUTPUT
DIS ₁	DIS ₂	I	$\bar{Y}$
L	L	L	H
L	L	H	L
X	H	X	(Z)
H	X	X	(Z)

FUNCTION TABLE—8T97

INPUTS		OUTPUT
DIS	I	Y
L	L	L
L	H	H
H	X	(Z)

FUNCTION TABLE—8T98

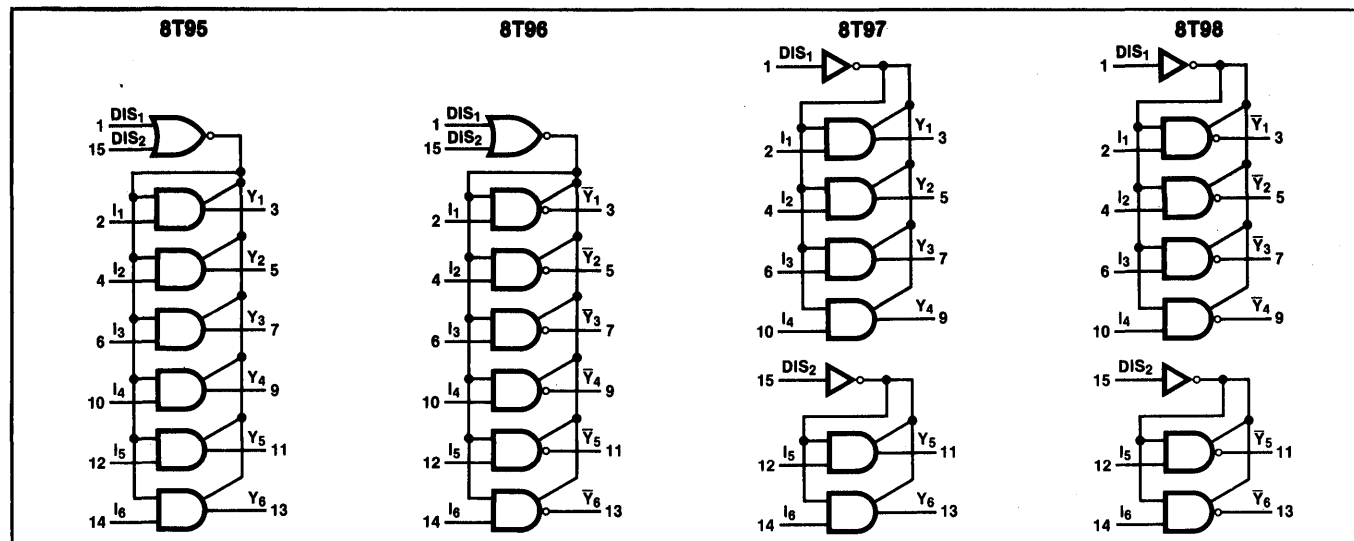
INPUTS		OUTPUT
DIS	I	$\bar{Y}$
L	L	H
L	H	L
H	X	(Z)

PIN CONFIGURATION

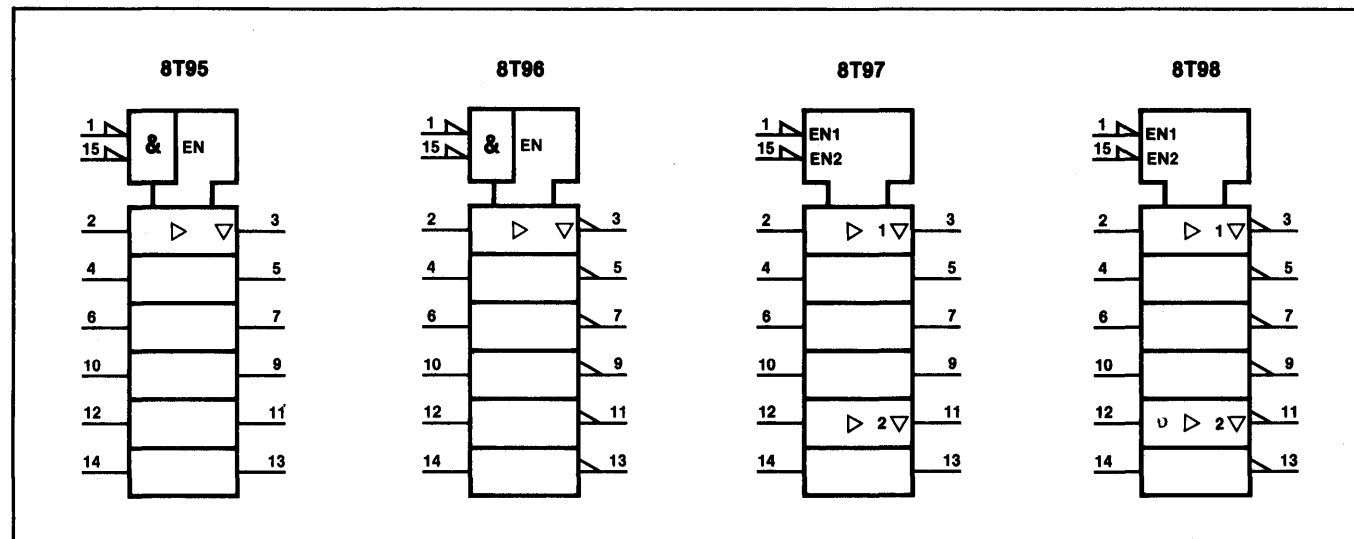
HEX BUFFERS/INVERTERS

8T95, 96, 97, 98

LOGIC SYMBOL



LOGIC SYMBOL (IEEE/IEC)



ABSOLUTE MAXIMUM RATINGS (Over operating free-air temperature range unless otherwise noted.)

PARAMETER	8T	N8T	UNIT
V _{CC} Supply voltage	7.0	7.0	V
V _{IN} Input voltage	-0.5 to +5.5	-0.5 to +5.5	V
I _{IN} Input current	-30 to +5	-30 to +5	mA
I _{OL} Continuous	100	100	mA
V _{OUT} Voltage applied to output in HIGH output state	-0.5 to +V _{CC}	-0.5 to +V _{CC}	V
T _A Operating free-air temperature range	-55 to +125	0 to 70	°C

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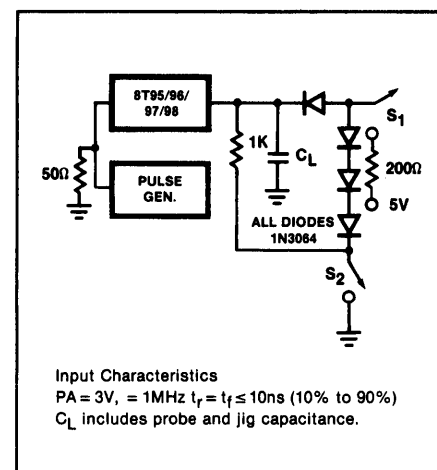
RECOMMENDED OPERATING CONDITIONS

PARAMETER		8T			UNIT
		Min	Nom	Max	
V_{CC} Supply voltage	Mil	4.5	5.0	5.5	V
	Com'l	4.75	5.0	5.25	V
V_{IH} HIGH-level input voltage		2.0			V
V_{IL} LOW-level input voltage	Mil			+ 0.8	V
	Com'l			+ 0.8	V
I_{IH} Input clamp current				- 18	mA
I_{OH} HIGH-level output current				- 5.2	μ A
I_{OL} LOW-level output current	Mil			48	mA
	Com'l			48	mA
T_A Operating free-air temperature	Mil	- 55		+ 125	$^{\circ}$ C
	Com'l	0		70	$^{\circ}$ C

NOTE

 $V_{IL} = +0.7V$ for S8T only.

TEST CIRCUIT



DC ELECTRICAL CHARACTERISTICS (Over recommended operating free-air temperature range, unless otherwise noted.)

PARAMETER	TEST CONDITIONS ¹	8T95/97		8T96/98		UNIT
		Min	Max	Min	Max	
V_{IH} Input HIGH voltage	Guaranteed input HIGH threshold voltage	2.0		2.0		V
V_{IL} Input LOW voltage	Guaranteed input LOW threshold voltage		0.8		0.8	V
V_{IK} Input clamp diode voltage	$V_{CC} = MIN$, $I_{IK} = -12mA$		-1.5		-1.5	V
V_{BD} Input breakdown voltage	$V_{CC} = MAX$, $I_I = 1mA$	5.5		5.5		V
V_{OH} HIGH-level output voltage	$V_{CC} = MIN$, $I_{OH} = -5.2mA$	2.4		2.4		V
V_{OL} LOW-level output voltage	$V_{CC} = MIN$, $I_{OL} = 48mA$		0.5 ³		0.5 ³	V
I_{OZH} Off-state output current, HIGH-level voltage applied	$V_{CC} = MAX$, $V_O = 2.4V$		40		40	μ A
I_{OZL} Off-state output current, LOW-level voltage applied	$V_{CC} = MAX$, $V_O = 0.5V$		-40		-40	μ A
I_{IH} HIGH-level input current	$V_{CC} = MAX$, $V_I = 2.4V$		40		40	μ A
I_{IL} LOW-level input current	$V_{CC} = MAX$, $V_I = 0.5V$	Disable = 0.5V	-400	Disable = 0.5V	-400	μ A
		Disable = 2.0V	-40	Disable = 2.0V	-40	μ A
I_{OS} Short-circuit output current ²	$V_{CC} = MAX$	-40	-115	-40	-115	mA
I_{CC} Supply current (total)	$V_{CC} = MAX$		98		89	mA

NOTES

1. For conditions shown as MIN or MAX, use the appropriate value specified under recommended operating conditions for the applicable type.

2. I_{OS} is tested with $V_{OUT} = +0.5V$ and $V_{CC} = V_{CC} MAX + 0.5V$. Not more than one output should be shorted at a time and duration of the short circuit should not exceed one second.3. $V_{OL} = +0.45V MAX$ for S8T at $T_A = +125^{\circ}C$ only.

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AC CHARACTERISTICS $T_A = 25^\circ\text{C}$, $V_{CC} = 5.0\text{V}$

PARAMETER	TEST CONDITIONS	8T95/97		8T96/98		UNIT
		R _L = 200Ω		R _L = 200Ω		
		Min	Max	Min	Max	
t _{PLH} Propagation delay Data inputs to Data outputs	S ₁ , S ₂ are closed, C _L = 50pF	3	12	4	11	ns
t _{PHL} Propagation delay Data inputs to Data outputs	S ₁ , S ₂ are closed, C _L = 50pF	3	13	3	10	ns
t _{PZH} Disable to outputs High Z to Logic “1”	S ₁ is open, S ₂ is closed, C _L = 50pF	8	25	7	22	ns
t _{PZL} Disable to outputs High Z to Logic “0”	S ₁ is closed, S ₂ is open, C _L = 50pF	12	25	11	24	ns
t _{PHZ} Disable to outputs Logic “1” to High Z	S ₁ , S ₂ are closed, C _L = 5pF	3	10	3	10	ns
t _{PLZ} Disable to outputs Logic “0” to High Z	S ₁ , S ₁ are closed, C _L = 5pF	3	12	5	16	ns

AC WAVEFORMS

